2025 28th International Conference on Computer and Information Technology (ICCIT)
19-21 December 2025, Cox's Bazar, Bangladesh

An Explainable Machine Learning Approach to Multiclass Diabetes Risk Assessment

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2025 28th International Conference on Computer and Information Technology (ICCIT) | 979-8-3315-7867-1/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICCIT68739.2025.11491480

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Abstract—Diabetes is a common and serious illness and the increasing number of cases puts pressure on both patients and the healthcare system. Early detection is essential, so we developed a model that predicts risk with higher accuracy. The study employs a full machine-learning pipeline. We cleaned the data by imputing missing values, identifying and removing outliers, and scaling the features. Since the classes were uneven, we applied SMOTE before training. Eight models were tested: Random Forest, XGBoost, Gradient boost, SVM with RBF, Logistic Regression, Na¨ıve Bayes, Extra Trees, and an ANN. The results were precise. Random Forest and XGBoost performed best, each achieving 98.11% accuracy. Their ROC-AUC scores were also very high, at 99.09% for Random Forest and 99.55% for XGBoost. The simpler models, such as Logistic Regression and Na¨ıve Bayes, remained much lower. We also added explainability to understand why the models behaved as they did. SHAP, PDP, and Tree Interpreter showed that HbA1c, BMI, and triglycerides played a significant role in the predictions. The framework retains the strong precision of the ensemble models while making decisions easier to interpret, which is helpful for fundamental clinical support tools.

Index Terms—Diabetes prediction, machine learning, ensemble models, explainable AI (XAI), clinical decision support

I. Introduction

Diabetes mellitus (DM) is now one of the most common long-term diseases in the world. Mostly, it involves problems with how the body processes glucose because insulin either does not function properly or the body does not produce enough of it. Reports from the International Diabetes Fed eration (IDF) show that more than 500 million adults live with diabetes today, and the number may increase even more in the coming years [1]. This increase imposes a substantial economic and health burden. Finding the disease early is essential because quick action can slow down serious issues such as heart disease, kidney damage, and nerve problems [2].

Doctors typically rely on basic clinical markers such as HbA1c, fasting glucose, and BMI. These markers help, but in many cases, they do not provide the whole picture. People come from different backgrounds, and risk factors do not behave in the same way between groups. As a result, standard methods may miss more complex patterns. Machine learning (ML) has become useful here. It can handle large datasets and

identify hidden links that are difficult to detect with simple statistical tests [3]. Stronger ML models, especially ensemble methods and deep networks, already show high performance in many health tasks, including diabetes prediction and mon itoring.

Even with these gains, one issue remains. Many ML models operate as black boxes. They present a result, but they do not explain how they obtained it. This makes doctors less comfortable using them in real care. As a result, explainable AI (XAI) tools are becoming increasingly important. Methods such as SHAP, PDP, and Tree Interpreter aim to identify the features that matter the most and how the model responds to changes in these features [5]. These explanations help bridge the gap between prediction and clinical understanding. In this study, we built a complete framework for multiclass diabetes detection. It uses both traditional ML models and an ANN. The process includes careful preprocessing, tuning, and testing. Subsequently, we apply XAI methods to clarify the results. The goal is simple: a model that is accurate, but also easy for clinicians to trust.

The study was carried out with a few clear goals. Each goal focuses on building a model that is both accurate and easy to understand in real clinical use.

1) Build a complete machine learning pipeline for mul ticlass diabetes prediction by cleaning the data set, handling imbalance, and testing a range of ML and ANN models.
2) To compare the performance of eight classifiers and identify which models yield the highest accuracy, F1- score, and ROC-AUC for this specific task.
3) Using explainable AI tools such as SHAP, PDP, and TreeInterpreter to demonstrate how key features such as HbA1c, BMI, and triglycerides influence predictions.

II. Related Work

In recent years, machine learning (ML) and deep learning (DL) approaches have been widely applied to the detection and prognosis of diabetes, producing promising results across diverse datasets. Numerous studies have explored supervised ML techniques, including Support Vector Machines (SVM), Random Forest (RF), and Logistic Regression (LR), to address

this issue for binary and multiclass diabetes classification. For example, Tigga and Garg [3] applied classical classifiers to the Pima Indians data set, achieving an accuracy of 81.0%, and the ensemble methods further improved performance. Ad ditionally, Gradient Boosting and Random Forest performed better than the older machine learning methods. Their models achieved approximately 85.7% accuracy, underscoring the strength of the ensemble approaches.

Deep learning has attracted attention because it can capture complex patterns that simple models miss. Alotaibi et al. [?] used an ANN and a CNN for the detection of type 2 diabetes and reached a precision of 89.3% %, with better sensitivity than many basic ML models. Rahman et al. [6] compared several deep learning frameworks and found that they scored above 90% precision and performed well in recall, especially for the identification of diabetic patients. Even with these gains, deep models still have a problem: it is difficult to explain how they make decisions, which reduces doctors confidence when using them.

Because many datasets are imbalanced, some researchers have attempted to resample to address this. Adeboye et al. [7] used SMOTE with Random Forest and increased balanced accuracy from 82.5% to 88.7%. Ali et al. [8] also employed a mix of SMOTE and ENN, achieving accuracies greater than 92.0% on multiclass diabetes tasks.

As trust became a greater concern, explainable AI began to receive more attention. Lundberg et al. [5] introduced SHAP into this area, allowing users to see both local and global characteristic effects while maintaining accuracy above 90%. Petch et al. [10] noted that interpretability is not optional in healthcare, particularly in high-risk settings such as diabetes diagnosis. Choi et al. [11] used PDPs to show how single features influence results, and other studies [13] used TreeIn terpreter to break down predictions into parts based on bias and features.

Some groups tried mixing ML with XAI from the start. Alshammari et al. [14] used XGBoost with SHAP to achieve a precision of 94.2%, again highlighting that HbA1c and BMI play an important role in the risk of diabetes. Kaur and Kumari [15] incorporated SHAP into an ANN and achieved 92.5%, providing clinicians with clearer insight into how the model works. Despite this progress, many studies still focus on accuracy or explainability, but not both. That gap motivates our work. We aim to build a framework that retains strong predictive power while being transparent via SHAP, PDP, and TreeInterpreter.

III. Methodology

The proposed framework for the prediction and inter pretability of diabetes is illustrated in Figure 1. The pipeline begins with *data preprocessing*, including handling missing values, outlier detection, and standard scaling to ensure high quality input. The data set is then split into training and testing subsets, with the training data further balanced using SMOTE to address class imbalance. Various machine learning models are trained and optimized through hyperparameter tuning, followed by a rigorous evaluation on the test set. Performance metrics and confusion matrices are recorded to identify the best-performing model. Finally, explainable artificial intelligence (XAI) techniques, including SHAP, Tree Interpreter, and Partial Dependence Plots (PDP), are applied to enhance interpretability and provide feature-level insight into model predictions.

A. Data Description

The data set used in this study is the *Multiclass Diabetes Dataset*, publicly available on Kaggle [16]. It comprises 1000 subjects, with 16 clinical and demographic features and a target variable indicating diabetes status. The data set supports multiclass classification with three outcome classes: non diabetic (0), diabetic (1), and prediabetic (2).

After preprocessing and removing duplicate entries, the data set contained 690 diabetic cases, 96 non-diabetic cases, and 40 prediabetic cases, resulting in considerable class imbalance.

The available features include: sex, Age, urine, Creatinine (Cr), HbA1c, Cholesterol, Triglycerides (TG), High-Density Lipoprotein (HDL), Low-Density Lipoprotein (LDL), Very Low-Density Lipoprotein (VLDL) and Body Mass Index (BMI). The target variable is labeled *Class*, with values: 0 = Non-Diabetic, 1 = Diabetic, and 2 = Prediabetic.

Among these features, HbA1c is a critical biomarker for the diagnosis of diabetes, whereas BMI and lipid profile indicators (HDL, LDL, VLDL, TG, Cholesterol) provide valuable insights into metabolic and cardiovascular health. The combination of biochemical, clinical, and demographic attributes makes the data set suitable for machine learning models aimed at predicting and adjusting diabetes risk.

B. Data Preprocessing

To ensure the quality and reliability of the dataset, several preprocessing steps were performed prior to the model devel opment. These include handling missing values, standardiza tion, and outlier detection/removal.

1) Missing Value Handling: The data set was examined for missing entries. Records with incomplete or inconsistent values were imputed using statistical measures (mean/median for continuous variables and mode for categorical variables). This process minimized information loss while preserving the size of the data set for training and evaluation.

2) Standardization: To bring all numerical features to a uniform scale and avoid bias towards variables with larger magnitudes, standardization was applied. Each continuous feature x was transformed using the z-score normalization:

$$z = \frac{x - \mu}{\sigma} \quad (1)$$

Here μ and σ denote the mean and standard deviation of the feature, respectively. This ensured that features such as Age, BMI, and biochemical indicators (e.g., Urea, Cholesterol, HbA1c) contributed equally during model training.

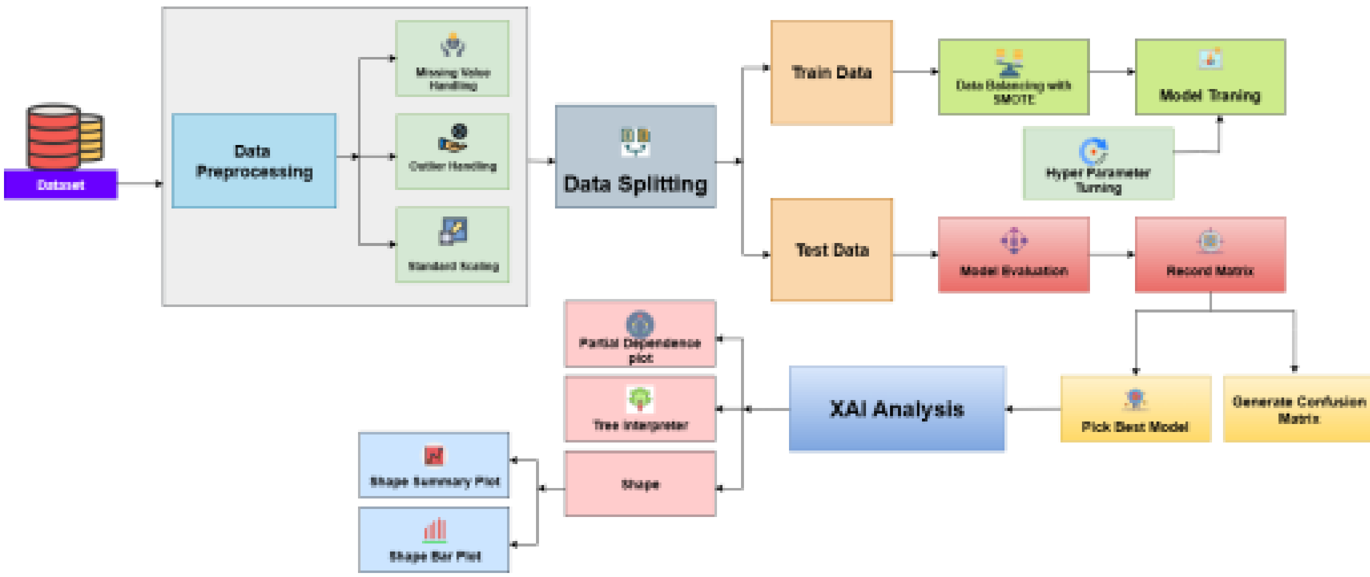


Fig. 1: Proposed methodology for diabetes prediction and explainability.

3) Outlier Handling: Outliers were detected and treated to prevent skewness in model performance. The Interquartile Range (IQR) method was employed, where the lower bound (LB) and upper bound (UB) are defined as:

$$LB = Q1 - 1.5 \times IQR, \; UB = Q3 + 1.5 \times IQR \quad (2)$$

Here, $Q1$ and $Q3$ represent the first and third quartiles, and $IQR = Q3 - Q1$. Any data points outside [LB, UB] were treated as outliers and capped or removed depending on their clinical plausibility.

C. Data Splitting and Balancing

1) Data Splitting: The preprocessed data set was divided into training and testing subsets to ensure an unbiased evaluation of the machine learning models. A stratified split was used to preserve the class distribution between the two subsets. Specifically, 80% of the data was allocated for training and 20% for testing.

2) Data Balancing with SMOTE: Due to the class imbal ance in the original data set (Diabetic = 690, Non-Diabetic = 96, Predict-Diabetic = 40), a Synthetic Minority Over sampling Technique (SMOTE) [9] was used. SMOTE gener ates synthetic data points for minority classes by interpolating between existing samples.

For a minority class sample x_i, one of its k nearest neighbors x_{nn} is randomly selected, and a synthetic point is generated as:

$$x_{new} = x_i + \delta \times (x_{nn} - x_i), \; \delta \sim U(0, 1) \quad (3)$$

learning models were considered to ensure robust performance evaluation. The selected models include:

- Logistic Regression (LR): A linear classifier that es timates class probabilities using the logistic (sigmoid) function. For a given input x, the probability of class c is expressed as:

$$P(y = c|x) = \frac{1}{1 + e^{-(\beta_0 + \beta^T x)}} \quad (4)$$

- Support Vector Machine (SVM): The objective is to maximize the margin between support vectors [4] while minimizing classification error. Its optimization problem can be written as:

$$\min_{w,b,\xi} \frac{1}{2} \| w \|^2 + C \sum_{i=1}^{n} \xi_i \quad (5)$$

subject to

where δ is a random number drawn from a uniform distribution $U(0, 1)$.

D. Model Development

To assess the predictive capacity of machine learning al gorithms for multi-class diabetes detection, several state-of-the-art classifiers were implemented. Traditional and deep

$$y_i(w^T \phi(x_i) + b) \geq 1 - \xi_i,\ \xi_i \geq 0 \quad (6)$$

where $\phi(x)$ maps the input data to a higher-dimensional feature space.

- Random Forest (RF): An ensemble of decision trees trained on bootstrapped samples. The final prediction is obtained by majority voting:

$$\hat{y} = \mathrm{mode}\{h_t(x)\}_{t=1}^{T} \quad (7)$$

where $h_t(x)$ is the prediction of the t-th tree.

- Extra Trees (ET): Similar to RF, but introduces more randomness by selecting split thresholds at random. The decision rule remains the same as Equation (4).
- Gradient Boosting (GB): Builds an additive model in a stage-wise manner. At each stage m, a new weak learner $h_m(x)$ is trained to minimize loss:

$$F_m(x) = F_{m-1}(x) + \eta h_m(x) \quad (8)$$

where η is the learning rate.

- Naïve Bayes (NB): A probabilistic classifier based on the Bayes' theorem with the assumption of conditional independence between features:

Logistic Regression and Naïve Bayes struggle to

$$P(y|x_1, x_2, \ldots, x_n) \propto P(y)\prod_{i=1}^{n} P(x_i|y) \quad (9)$$

- XGBoost (XGB): An advanced gradient boosting method [12] that optimizes the following objective:

prediction. Random Forest and XGBoost reached the

$$L(\theta) = \sum_i l(y_i, \hat{y}_i) + \sum_k \Omega(f_k) \quad (10)$$

F1-Score stayed above

where l is a differentiable convex loss function and $\Omega(f_k)$ is a regularization term to control the complexity of the model.

- Artificial Neural Network (ANN): A deep learning model consisting of multiple layers. Each hidden layer computes:

$$h^{(l)} = f(W^{(l)}h^{(l-1)} + b^{(l)}) \quad (11)$$

where $f(\cdot)$ is the activation function. The final output layer uses the softmax function for multi-class classifica

cholesterol levels, and BMI.

tion:

$$\hat{y}_i = \frac{e^{z_i}}{\sum_{j=1}^{K} e^{z_j}},\ i = 1, 2, \ldots, K \quad (12)$$

where z_i is the logit of class i and K is the number of classes.

All models were trained using stratified training data (with SMOTE balancing) and validated on unseen test data. Hyper parameter tuning was conducted using grid search with cross validation to optimize model performance.

IV. Results

The experimental evaluation was conducted on the multi class diabetes data set after applying preprocessing, SMOTE balancing, and 80:20 train–test splitting. The models were as sessed using Accuracy, Precision, Recall, F1-Score (Weighted and Macro), and ROC AUC.

Table I presents the comparative results for seven ma chine learning models, including an Artificial Neural Network (ANN). Random Forest and XGBoost achieved the best overall performance, each with an Accuracy of 98.11%, a Precision (weighed) of 98.21%, a Recall (weighed) of 98.11% and an ROC AUC exceeding 99%. Gradient Boosting followed with 96.23% Accuracy and 96.23% weighted F1. Extra Trees

differentiate Predict-Diabetic samples, often misclassifying them as Non Diabetic or Diabetic.

V. Discussion

The results indicate that ensemble-based models outper form traditional classifiers and ANN in multiclass diabetes

highest accuracy in the study, and their macro Precision, Recall, and

98%. These results show that both models can handle the imbalance of data and still learn the complex associations among features such as HbA1c,

The confusion matrices make one point clear. The hardest group to predict is the Predict-Diabetic class. Random Forest and XGBoost handled it well, with very few errors, whereas the simpler models did not. Logistic Regression and Naïve Bayes often mixed up these cases and labeled them as Diabetic or Non-Diabetic. In real use, this kind of mistake matters because a wrong label here can slow down early care. The ANN model performed well, achieving 94.34% accuracy and a strong ROC AUC of 98.72%. However, it could not match the two top models. It seems that the network can learn some deeper patterns, but a larger or more tuned version would likely be needed to achieve the same level of general performance. In general, the findings establish that ensemble learning, supported by proper data balancing (SMOTE), provides a robust framework for the prediction of diabetes. Future ex tensions could integrate deep learning hybrid ensemble strate gies and domain-specific feature selection to further enhance sensitivity to minority classes.

VI. Explainable AI (XAI)

To ensure model transparency and clinical interpretability, we analyzed the best-performing model (Random Forest) with two complementary post-hoc explainability techniques: SHAP (Shapley Additive exPlanations) and Partial Dependence Plots (PDP). SHAP provides both global and local attributions grounded in cooperative game theory, while PDP visualizes the marginal effect of a feature on the predicted probability for each class.

A. SHAP: Global and Local Feature Attributions SHAP explains a prediction by distributing the difference between the model output and a baseline among the input features. For a model *f* and a set of characteristics *S*, the Shapley value for the characteristic *i* is

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!\,(|F| - |S| - 1)!}{|F|!} \left[f(S \cup \{i\}) - f(S) \right], \quad (13)$$

where *F* is the complete feature set and *f*(*S*) denotes the model output when only the features of *S* are present. Figure 3 summarizes SHAP results. The bar plot (left) ranks features by mean absolute SHAP value (global importance), while the beeswarm plot (right) shows per-sample signed impacts. In both views, HbA1c, BMI and Age emerge as the dominant predictors, aligning with clinical knowledge.

and ANN achieved competitive accuracies of 94.34%. In contrast, Logistic Regression and Naïve Bayes recorded a lower performance, with Logistic Regression achieving only 86.79% Accuracy and 87.07% weighted F1.

Figure 2 shows the confusion matrices of the models. Random Forest and XGBoost exhibit strong predictive power across all three classes (Non-Diabetic, Diabetic, Predict Diabetic), with very few misclassifications. Gradient Boosting and ANN also demonstrate consistent performance, but with slightly more misclassifications in the Predict-Diabetic class.

TABLE I: Performance comparison of ML models and ANN. Metrics used: Accuracy, Precision (W = Weighted, Macro), Recall (W = Weighted, Macro), F1-Score (W = Weighted, M = Macro), and ROC AUC.

Model	Accuracy	Precision (W)	Recall (W)	F1-Score (W)	Precision (M)	Recall (M)	F1-Score (M)	ROC AUC
RandomForest	98.11	98.21	98.11	98.12	98.33	98.72	98.49	99.09
XGBoost	98.11	98.21	98.11	98.12	98.33	98.72	98.49	99.55
GradientBoosting	96.23	96.53	96.23	96.23	94.63	97.44	95.85	99.62
ExtraTrees	94.34	94.34	94.34	94.34	92.80	92.80	92.80	99.77
SVM (RBF)	92.45	93.99	92.45	92.66	89.63	93.45	90.96	98.65
Naïve Bayes	90.57	92.09	90.57	90.47	94.62	88.82	90.93	98.03
Logistic Regression	86.79	88.09	86.79	87.07	82.11	83.37	82.24	94.27
ANN	94.34	94.76	94.34	94.48	91.31	92.81	91.94	98.72

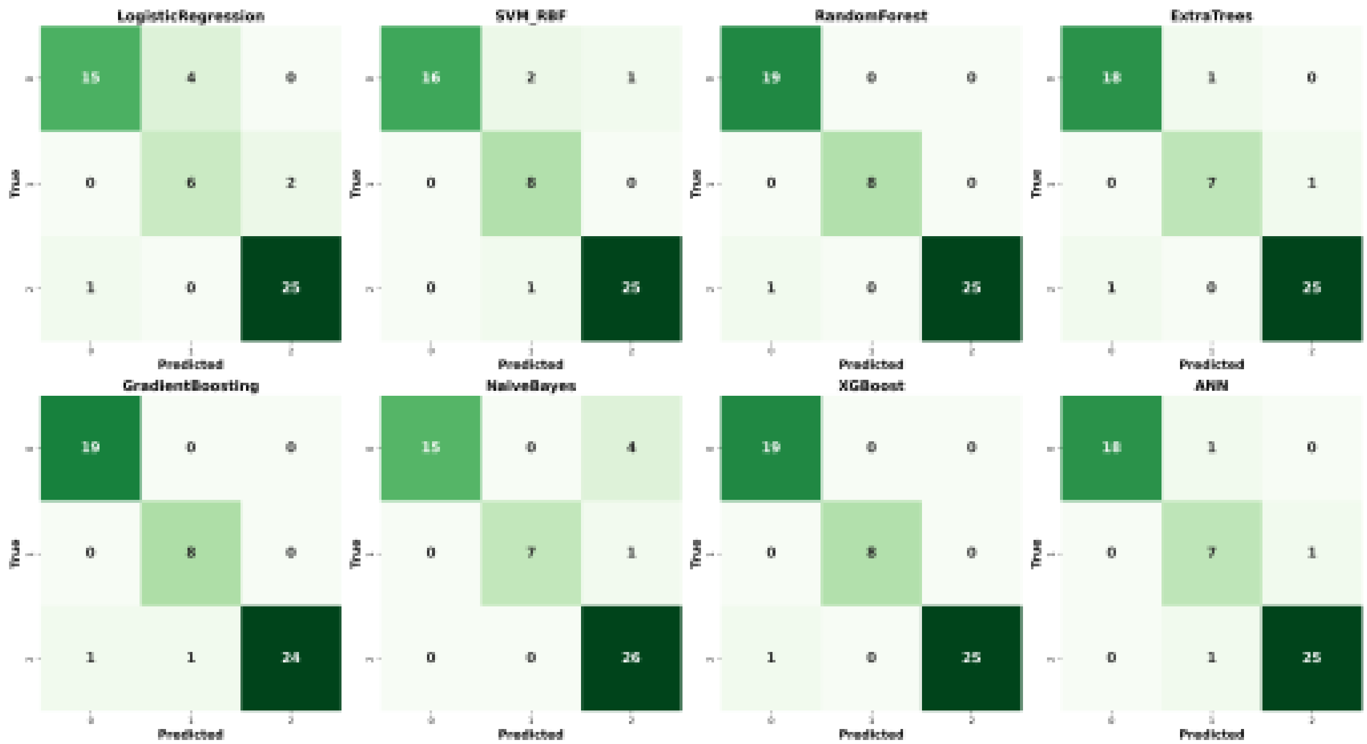


Fig. 2: Confusion matrices of different models on the multiclass diabetes dataset. The diagonal values represent correctly classified samples, while off-diagonal entries indicate misclassifications.

B. Partial Dependence Plots (PDP)

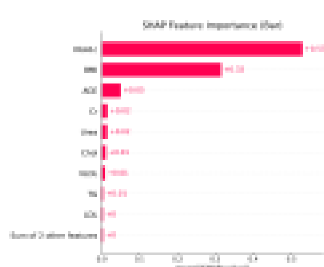
PDPs depict the marginal effect of a characteristic x_j on the output of the model by averaging the empirical distribution of all other features $X_{\setminus j}$:

$$\mathrm{PDP}_j(z) = \mathrm{E}_{X_{\setminus j}}\, f(z, X_{\setminus j}) \,. \quad (14)$$

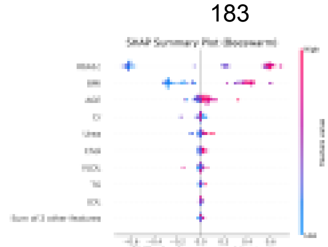
For the multiclass setting, we plot the class-wise partial dependence on the predicted probability for each class. Figure 4 aggregates the PDPs for the top three features (HbA1c, BMI, Age) in all three classes. The curves indicate *monotonic* or *threshold-like* responses consistent with SHAP: increasing HbA1c generally increases the probability of the diabetic class; BMI exhibits a positive association with higher risk classes; and Age shows a moderate, class-dependent effect.

C. Interpretation Summary

SHAP and PDP jointly confirm that glycemic control (HbA1c), adiposity (BMI), and Age drive the model's de cisions. SHAP provides case-level attributions appropriate for patient-specific explanations, while PDP offers global response curves that reveal marginal trends (e.g., risk escalation with higher HbA1c). The congruence between these methods sup ports both the *predictive validity* and *clinical plausibility* of the learned patterns.



(a) Global importance (mean |SHAP|).



(b) Sample-level impacts (signed SHAP values).

lipid markers stood out in the samples and matched what we see in real clinical practice. The explainable AI tools also made the model less like a black box, which is necessary if doctors are going to trust it. However, the study used a limited dataset, so the approach needs to be tested in larger, more diverse groups before it can be used with confidence everywhere. A possible next step is to implement this system in a real healthcare setting, where it can run in the background, support early checks, and help clinicians make decisions directly and straightforwardly.

Fig. 3: SHAP-based explainability of the Random Forest model. High positive SHAP values push predictions toward the corresponding class, while negative values

push away.

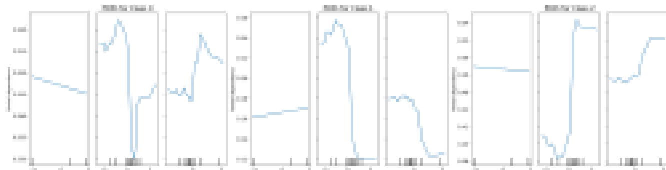


Fig. 4: Partial Dependence Plots for the Random Forest model showing class-wise probability response to HbA1c, BMI, and Age. Each panel reflects the marginal effect while averaging over the remaining features.

D. TreeInterpreter: Instance-Level Decomposition

To further complement SHAP and PDP, we employed the treeinterpreter framework to analyze feature contri butions at the instance level. The prediction of the model for a representative test case was [0.9833, 0.0100, 0.0067], strongly favoring Class 0. The bias (intercept) terms were approximately uniform across classes (≈ 0.33), and the final prediction was obtained by adding individual feature contribu tions to this baseline. Among the features, HbA1c (+0.3831), BMI (+0.0884), and Triglycerides (+0.0646) emerged as the strongest positive drivers toward the predicted class, fol lowed by moderate effects from Age (+0.0338) and VLDL (+0.0334). In contrast, HDL (–0.0001) and LDL (–0.0000) had negligible influence. These findings align closely with the SHAP global importance rankings, reinforcing the critical role of HbA1c and BMI in disease classification. Thus, the tree interpreter provides a transparent breakdown of how individual features jointly steer the model's decision for a specific patient case.

VII. Conclusion

This work demonstrates the potential of machine learning to effectively predict diabetes, inspiring confidence among medical professionals in AI's role in healthcare. The ensemble models, primarily Random Forest and XGBoost, achieved the highest scores and performed nearly flawlessly in many tests. Adding SHAP and PDP values helped us identify which features were the most informative. HbA1c, BMI, and several

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